VEIL

An adversarial privacy auditor that attacks your document before someone else does.

Fully local inference · vLLM on AMD ROCm · Zero egress

Private AI Agents · AMD Track

THE FAILURE MODE

Redaction fails constantly — and silently.

In December 2025, a US government release of court records shipped with redactions that could be undone by copy-paste. Names of victims and witnesses were exposed.

That was not an unusual event. It is the normal outcome of a manual process running at roughly five minutes per page under deadline.

95%

of 2024 breach events traced to human error

~5 min

manual review cost per page, per pass

€20M

or 4% of global revenue — GDPR maximum penalty

WHY TODAY'S TOOLS MISS IT

Pattern matchers find patterns. Re-identification isn't a pattern.

WHAT PRESIDIO / COMPREHEND PII SEE

Strings shaped like names, emails, phone numbers

Regexes and a fixed entity list

Each span judged in isolation

No notion of who a combination points to

Verdict on our demo document: no issues found

WHAT ACTUALLY DEANONYMISES PEOPLE

Text still extractable under a drawn redaction box

Quasi-identifiers that intersect to one person

Role, tenure and reporting-line inference

Authorship style; document metadata remnants

VEIL recovers the identity in under a minute

THE ARCHITECTURAL ARGUMENT

This cannot be a cloud product.

01

The document is the secret

You cannot upload a sealed filing, a clinical record or a source's testimony to a third-party API to check whether it is safe to release. The check would be the disclosure.

02

Frontier APIs refuse the task

Asking a hosted model to de-anonymise a person is a policy violation. The one capability we need is the one they are aligned to withhold.

03

Open weights on local silicon solve both

A quantised open model on a Radeon GPU will run the attack, and the bytes never leave the box. Local inference is not a constraint we tolerate — it is the only way this product can exist.

HOW IT WORKS

Attack. Adjudicate. Repair. Re-attack.

One vLLM server. Five adversarial personas running concurrently against the same document. A Guardian that proposes the smallest edit that breaks each attack.

Ingest

PDF · DOCX · TXT
normalised to spans

Adversary Pool

5 personas, concurrent
shared prompt prefix

Adjudicator

dedupe · tier
severity · candidate set

Guardian

minimal typed edits
suppress / generalise

Verify

re-attack the fix
until every persona fails

The loop is the product

A finding is only marked resolved when every persona re-runs against the edited document and fails to reproduce it. VEIL proves its own fix — no human is asked to take its word.

THE ADVERSARY POOL

Five attackers, five threat models, one batched GPU pass.

Layout-Forensicist

Redaction-box geometry, PDF metadata, revision remnants, character-width leaks.

TIER M · VERIFIABLE

Data-Broker

Intersects quasi-identifiers — dates, geography, role — and reports the candidate-set size.

TIER C · K-ANONYMITY

Insider

Assumes org-chart knowledge. Targets tenure, reporting lines and role leakage.

TIER C · CORRELATIONAL

Journalist

Open-source correlation against the auxiliary corpus. Who is publicly known to match?

TIER C · CORRELATIONAL

Stylometrist

Authorship attribution from idiolect and punctuation habit. Always hedged.

TIER S · LOW CONFIDENCE

Why five?

Concurrency saturates vLLM continuous batching. The persona count is set from a measured throughput sweep, not a guess.

CALIBRATED OUTPUT

Three evidence tiers. No overclaiming.

A privacy tool that cries wolf gets switched off. Every finding is typed by how much it can actually be trusted, and the interface renders the difference.

M

MECHANICAL

Directly extractable. Certain.

"The text beneath the black box on page 3 is Dr. Amara Okonkwo."
Ships with a Verify button that shows the raw extraction.

C

CORRELATIONAL

Inferred. Quantified.

"Regional office + Q2 2019 hire date + ASL certification narrows to an estimated 3 individuals."
Always carries a candidate count.

S

STYLISTIC

Suggestive. Never actioned alone.

"Phrasing is consistent with the author of Exhibit 7 — style attribution on short texts is unreliable." Explicit caveat attached.

Every report closes with a residual-risk statement: what we did not check, and what an attacker with different auxiliary data might still find.

OPERATOR CONTROL

The privacy/utility tradeoff, made visible.

Naive redaction destroys the document. The Guardian proposes the minimal edit — generalise a date, broaden a certification, suppress one span — and every iteration plots a point on the curve.

Publish

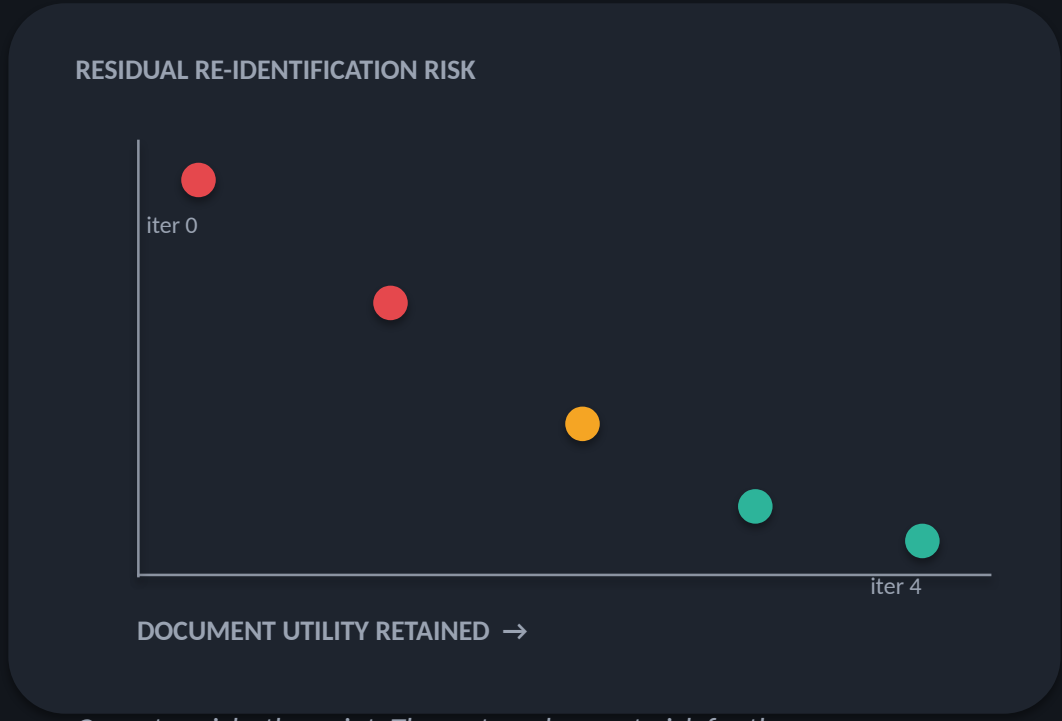
Aggressive. Assumes a hostile, well-resourced reader.

Legal Disclosure

Balanced. Preserves evidentiary value.

Source Protection

Maximum. Accepts heavy utility loss.



Operator picks the point. The system does not pick for them.

ARCHITECTURE

One model, one server, five roles.

PRESENTATION	Single-page UI via rc-tunnel · session-token auth · streaming attack theatre
APPLICATION	FastAPI · async orchestrator · WebSocket event bus · role gate
AGENTS	Planner · Adversary Pool (5) · Adjudicator · Guardian
TOOLS	layout_forensics · doc_metadata · corpus_search · candidate_set · entity_ledger · redact_apply
DATA	Qdrant vectors · SQLite findings · Fernet-encrypted blobs · hash-chained audit log
INFERENCE	vLLM on ROCm · Qwen2.5-14B-Instruct-AWQ · prefix caching · continuous batching

No LangChain, no build step, no second model. Framework debugging is the single largest time sink in a 24-hour build.

AMD OPTIMISATION

Prefix caching is what makes five attackers cost barely more than one.

DETERMINISTIC PROMPT CONSTRUCTION

SHARED PREFIX — byte-identical across every persona and iteration

```
system preamble
document text  ← the long part
extracted spans + entity ledger
```

PERSONA SUFFIX — varies

```
role · attack instruction · tools · prior findings
```

5 personas × 3 iterations → document KV computed once, reused 14×.

`--enable-prefix-caching`

The single most consequential flag in the build.

`--quantization awq --dtype float16`

AWQ is the reliable path on ROCm. GPTQ is unstable; GGUF is unsupported by vLLM. Dtype is pinned, never left to auto.

`--max-num-seqs 16`

Set from a measured concurrency sweep. Too low and concurrency becomes queueing, not batching.

Fallback ladder, decided hour 1

gpu-mem 0.90 → 0.85 → max-len 8192 → Qwen2.5-7B-AWQ. Rehearsed, not improvised at hour 20.

MEASUREMENT PLAN

What we instrument, and how.

Hours 14–16 of the build are a dedicated benchmark block. Numbers below are captured on the real demo path — not a synthetic loop — and go in the README.

METRIC	HOW	WHY IT MATTERS
Tokens/sec, single stream	`bench/benchmark.py --concurrency 1`	baseline
Tokens/sec, 5 concurrent personas	`--concurrency 5`	batching gain
Concurrency sweep 1 → 16	`--sweep`	locates the knee; justifies N=5
Prefix cache hit rate	`vLLM /metrics`	validates the core optimisation
GPU utilisation %, 1 Hz trace	`rocm-smi --showuse`	headline AMD number
VRAM high-water mark	`rocm-smi --showmemuse`	proves an engineered budget
End-to-end audit wall time	`app timer`	the number a customer asks about

A GPU utilisation trace plotted against the attack timeline visibly shows the card saturating as the persona pool fans out.

TRUST ARCHITECTURE

Our own output is the most dangerous thing in the building.

A VEIL findings file is a working re-identification attack report. The system is designed on that assumption.

Zero egress, enforced

An allowlist HTTP client permits only localhost vLLM and Qdrant. A startup self-test attempts an external call and displays the result. The UI carries a live counter: 0 external calls this session.

Role separation

AUDITOR sees raw findings and evidence. REDACTOR sees only proposed edits and the diff. VIEWER sees the final document and the residual-risk statement — nothing more.

Hash-chained audit log

Every load, tool call, model call, finding, edit and export writes a record whose hash includes its predecessor. Tampering is detectable, and the UI verifies the chain live.

Encrypted at rest, purged by default

Findings are Fernet-encrypted under a PBKDF2-derived key that never touches disk. Document text and findings are purged at session end unless explicitly exported.

SEVEN MINUTES, LIVE

The demo judges can verify themselves.

0:00

Presidio finds nothing

We run the industry baseline live on a failed-redaction PDF. It reports no issues.

1:15

VEIL recovers the name

The Layout-Forensicist returns a Tier M finding in seconds. Click Verify — raw extraction shown.

2:15

Ground truth revealed

On a synthetic-org document, quasi-identifier intersection narrows to one person. We open the answer key.

3:30

Live rocm-smi trace

GPU utilisation jumps as the pool fans out. Benchmark table and cache hit rate shown.

4:15

Guardian fixes it

Minimal redline. Re-attack. Every persona now fails; candidate set goes from 1 to 40+.

6:00

Trust, then limits

Verify the audit chain live, switch to REDACTOR role, then name our honest failure case.

Anti-scripting guarantee: a live paste box lets judges type their own text and watch it get attacked.

24 HOURS, THREE PEOPLE

Scoped to finish, then rehearsed.

H 0–8

Serve, ingest, fan out

Model on PVC and vLLM answering by hour 2. Five personas running concurrently by hour 8.

H 8–14

Adjudicate and repair

Tiering, severity, the tool registry, Guardian edits and the verification loop.

H 14–18

Benchmark and harden

Dedicated benchmark block, security layer, ≥40 tests, README with real numbers.

H 18–24

Freeze and rehearse

Hard feature freeze at hour 18. Bug-bash the demo path only, then four full rehearsals.

MUST EXIST AT HOUR 18

- Live concurrent attack with a Tier M finding Presidio misses
- Guardian fix verified by re-attack
- Benchmark table with real ROCm numbers

EXPLICITLY DEFERRED

- OCR for scanned documents — stubbed with a visible flag
- Multilingual support; cross-document correlation UI
- Signed audit-certificate export

WHERE THIS GOES

Assurance, not replacement.

VEIL runs after your existing redaction process and before publication. It does not ask anyone to change how they work — it tells them whether it worked.

Litigation support

Sealed filings and discovery productions. Sanctions and malpractice exposure make the cost of a miss enormous, and review is already billed by the page.

Clinical research

HIPAA Expert Determination currently rests on statistical judgement. VEIL supplies adversarial evidence for the determination file.

Investigative journalism

Source protection where the newsroom cannot upload the document anywhere. Local-only is the entire requirement.

The defensible moat

The attack is only as good as the auxiliary corpus behind it — and the corpus that matters is the customer's own archive, which is exactly what they cannot send to a cloud vendor. Corpus building is a first-class product feature, and it compounds.

VEIL

Find out what your document gives away while it is still yours to fix.

Local

core inference on Radeon, zero external calls

Concurrent

five personas, one server, shared prefix cache

Verified

every fix re-attacked before it is accepted